

FIELD REPORT

VULNERABILITY IN AFRICAN SMALL-SCALE FISHING COMMUNITIES

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Abstract: Fishing communities are often recognised as being amongst the poorest in developing countries, and interventions targeted at improving resource status seen as central in the fight against poverty. A series of field assessments focusing on vulnerability conducted in two communities in Mali and Nigeria revealed some counterintuitive results. Despite fishing being the primary livelihood, vulnerabilities relating directly to the state of the resource were ranked lower than those relating to basic human needs. Those results challenge the conventional view and suggest that non-sectoral interventions can have more effective impacts on the livelihood of those communities than interventions targeting the resources. Copyright © 2009 John Wiley & Sons, Ltd.

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1 INTRODUCTION

Small-scale fishing communities are often characterised as being amongst the most destitute socio-economic groups in developing countries (Jazairy *et al.*, 1992). This view, which is commonly accepted in the fisheries literature (see Béné, 2003 for a review), conveys the idea of a structural, chronic poverty. Other authors highlight instead the high

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vulnerability of fisherfolks, partially due to their high exposure to certain natural, health-related or economic shocks and disasters (Allison *et al.*, 2006; Allison *et al.*, 2009; Béné, 2009).

The project described here aims at addressing those issues of poverty and vulnerability in fishing-dependent communities in sub-Saharan Africa. Two pilot sites were selected to implement this research, one in the Inner Delta of the Niger River in Mali and one on the shore of the Lake Kainji in Nigeria. In those regions, populations depend on a combination of activities to sustain their livelihoods, amongst which fishing, but also farming, and livestock raising constitute the main components.

The first step in this research was to identify the sources of vulnerability affecting the livelihoods of these communities. This field report presents the results of this initial analysis and discusses the potential implications in term of future interventions.

2 METHODS

Vulnerability analyses were implemented in the two pilot sites using participatory assessment techniques, with the objective to identify key threats and external drivers affecting the communities' livelihood. Forty households in the Nigeria community and ninety in the Malian one were selected randomly. The two main adults (man and woman) of the households were interviewed separately in each household. A 360° integrated assessment map (Figure 1) was used to explore the various constraints affecting those households' livelihoods. Based on this, individuals were then asked to rank what they perceived as being their main sources of vulnerability.

In addition, a socio-economic questionnaire was administered in each household. The questionnaire included quantitative semi-open and closed (multi-choice) questions. The questionnaire was structured to collect detailed information on household income, general socio-economic background (number of persons, ethnic group, age, education), productive assets and livelihood strategies (on-farm and non-farm activities). Income referred to the household's income earned in cash plus payment in kind that could be valued at market

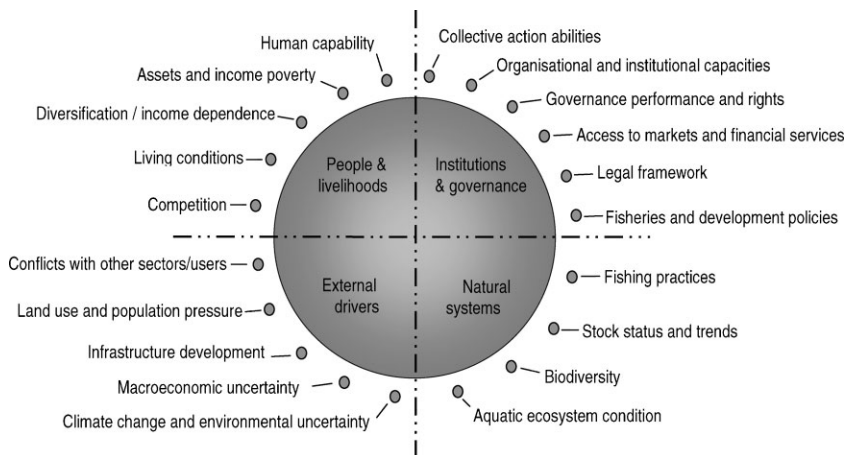


Figure 1. The 360° integrated assessment map used to structure the vulnerability assessment. The objective is to 'scan' in a systematic and comprehensive manner the household context in order to gain a better appreciation of the true nature of drivers and processes that affect livelihoods

prices. The cash-earning components of income included crop and vegetable sales, petty trade, remittance and fish sales.

It was hypothesised that the sources and intensities of vulnerability affecting the different households would depend on, or at least reflect, the main economic activity of the households. Similarly, it was hypothesised that wealth would also influence the household vulnerability structure.

3 ANALYSIS/RESULTS

Vulnerability ladders¹ developed from survey responses (Figure 2) present some important similarities between the two communities. In particular, in both communities, food insecurity and health issues rank amongst the three most important sources of vulnerability. Lack of cash and poor access to capital was also identified as a major issue. This in itself is not surprising as poor access to (formal) micro-credit has been long recognised in rural

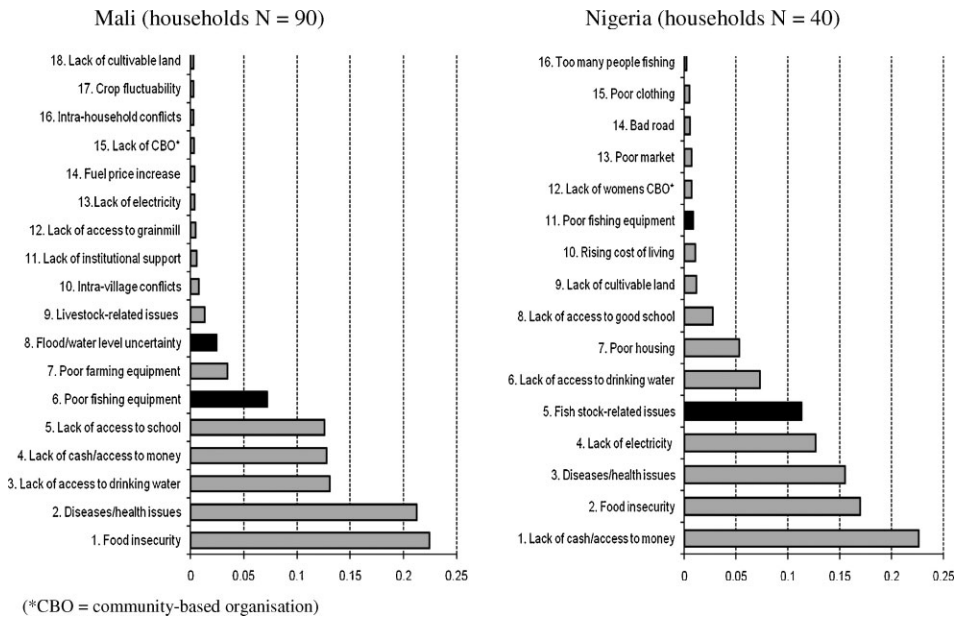


Figure 2. Vulnerability ladders of the two fishing communities surveyed. In black are indicated the sources of vulnerability related to the fish stock and/or fishing activities

¹These vulnerability ladders were computed by aggregating the individual households' answers and normalising the scores (total = 1) as follows: Let V_i be a source of vulnerability (e.g. V_i = 'food insecurity'), $i = 1$ to n , where n = total number of sources of vulnerability identified by the households in the community (e.g. $n = 16$ in the Nigerian community). Respondents were asked to identify the first five main sources of vulnerability for their individual household. For each source of vulnerability V_i a vulnerability score k_{Vi} across the community was then computed as follows: $k_{Vi} = \sum_{j=1}^5 \alpha_j \times N_j$ where N_j is the number of times (counts) the source of vulnerability V_i was mentioned by households in round j , ($j = 1-5$), and α_j is a weighting vector. In our case $\alpha_j = (1; 0.95; 0.9; 0.75; 0.33)$, which means that vulnerability sources identified as first main sources are weighted 1, 2nd sources were weighted 0.95, 3rd source: 0.9, and so forth. The vulnerability scores were then normalised: $k_{Vi}^* = \frac{k_{Vi}}{\sum_{i=1}^n k_{Vi}}$ so that $\sum_{i=1}^n k_{Vi}^* = 1$ and direct comparisons between villages and between sub-samples were possible.

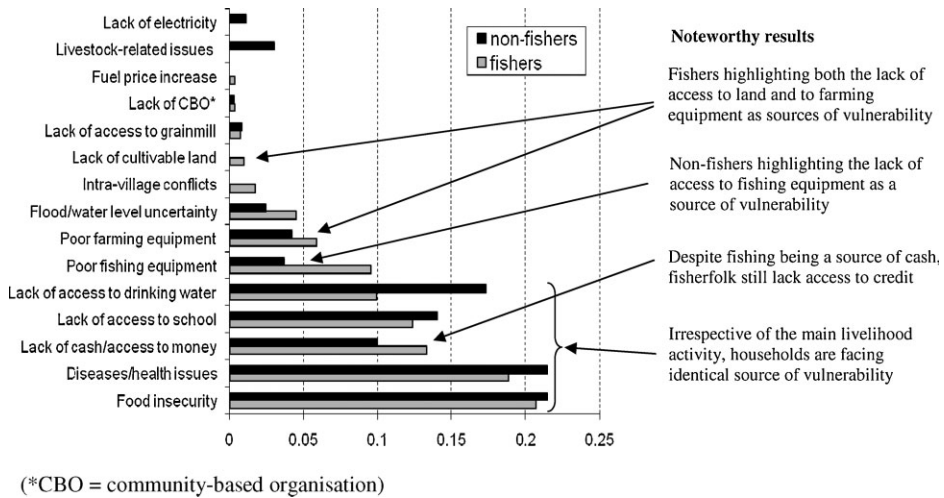


Figure 3. Comparative analysis of vulnerability between fishers and non-fishers in the Inner Delta of Niger River (Mali). (fishers $N=25$; non-fishers $N=30$)

development literature as one of the major constraints for poverty alleviation (IFAD, 2001; World Bank, 2002). Perhaps more surprising is that in both communities, fishing activities and fish resources issues (highlighted in black in Figure 2) are given a relatively low ranking.

Clearly it cannot be assumed that the sources of vulnerability identified in the survey are independent of one another. Indeed, it could be argued that the importance of fishing activity as a major source of cash-income and food transpires indirectly in the ladders through the presence of 'food insecurity' and 'lack of cash/access to money' at the top of the ranking². An analysis of vulnerabilities in the Malian community disaggregated by main livelihood activities (fishers and non-fishers³) (Figure 3) suggests however that this assertion may be only partially correct. The vulnerability ladder indicates that the sources of household vulnerability remain essentially the same, irrespective of the households' main livelihood. In other words, 'food insecurity' and 'lack of cash/access to money' may not be related to the fact that households are fishers, but rather represent overarching vulnerabilities that affect the community as a whole.

The comparative analysis between fishers and non-fishers reveals other policy-relevant results (see Figure 3 annotations). First, that scarcity of cultivable land is identified as a source of vulnerability, not by the non-fishers (farmers), but by the fishers, suggests that farming would be considered as a source of diversification by those fishers if they could access more land. This interest in farming is confirmed by the fact that 'poor farming equipment' is also perceived as a more severe constraint by fishers than by the non-fishing households. Less surprisingly, 'poor fishing equipment' affects fishers. The fact that non-fishers also identify this lack of fishing equipment as a source of vulnerability, however, suggests that non-fishers would also invest in fishing if they could access good equipment. Finally, the higher severity of the 'lack of cash/access to money' amongst fishers (compared

²One may even argue that the 'disease/health issue' (which refers not only to a lack of access to health facilities but also to the household inability to pay for those services), is also related to that general lack of cash, thus possibly related to fishing activities and the poor resource status.

³'Fishers' were defined as households who derive 75 per cent or more of their total income from fish-related activities while 'non-fishers' are those who derive no cash-income from fishing (essentially farmers and herders).

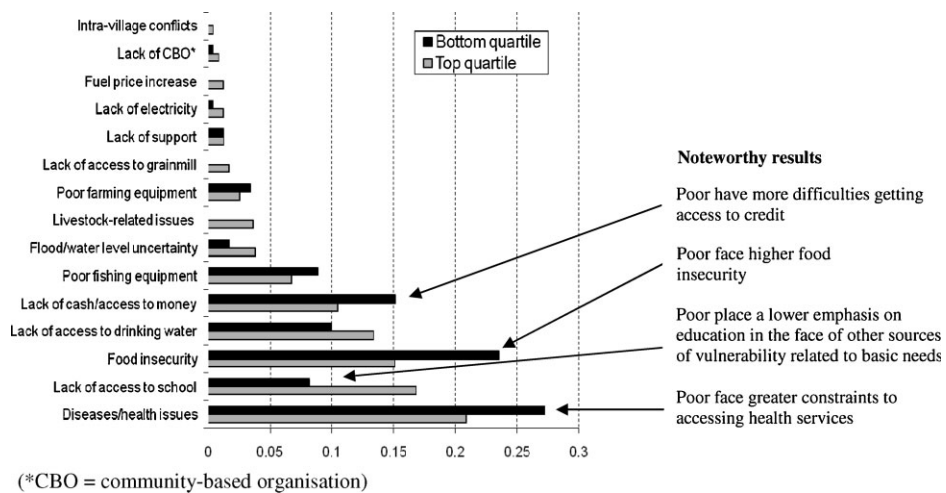


Figure 4. Comparative analysis of household vulnerability between the top (richest) and bottom (poorest) quartiles for the Inner Delta (Mali) community

to non-fishers), despite the well-recognised capacity of fishing to generate cash (e.g. Béné *et al.*, 2009), could possibly be related to the unwillingness of the local money lenders to provide credit to migratory fishers who are more difficult to track for repayment than the sedentary farmers.

Comparing vulnerability rankings among the bottom (poorest) and top (richest) income quartiles at the Malian pilot site⁴ (Figure 4) unsurprisingly shows that the poorest households are more severely affected by the lack of cash, food insecurity and health issues than the wealthier households. Also interesting (but not surprising) is that education is superseded by more critical 'life support' categories for the lowest income quartile. Aside from the ranking of education, the analysis reveals that households, independent of wealth, are facing the same sources of vulnerability.

4 CONCLUSION

While providing a valuable self-assessment of the priorities for reducing vulnerability in target communities, these results also contain important insights into how poverty/vulnerability interventions in fishing communities should be conceived; they certainly challenge the conventional view that efforts should primarily focus on the resource. Although fish resource depletions/fluctuations are acknowledged and certainly affect their livelihood, the communities identified some more fundamental sources of vulnerability related to their basic needs, such as food insecurity, exposure to water-borne diseases and lack of access to cash and micro-credit facilities.

The vulnerability rankings also convey important lessons for resource managers charged with implementing sustainable use systems under the constraints of minimal resourcing—a common condition among small-scale fisheries in developing countries. Under such constraints, private incentives for collective action among resource users are critical to ensure successful and equitable management outcomes (Olson, 1965; Ostrom, 1990; Thomson,

⁴Similar results are observable in the Nigerian community. Those are not presented here due to lack of space.

1992; Vedeld, 2000). Incentives for individuals to invest in resource sustainability will increase as vulnerabilities ranked higher by the community are addressed, and the risk of fishery decline comes to the fore. It follows that the most productive interventions to promote sustainable resource use may lie outside the 'natural system' domain (see also Béné and Friend, 2009); in this case within the 'people and livelihoods' (for water-borne diseases) and 'institutions and governance' (for access to micro-credit) domains (see Figure 1).

Outcomes of this preliminary analysis emphasise the value of participatory, multi-sectoral diagnosis. Action-research components of the programme currently implemented aim to facilitate networks of local institutions to support cross-sectoral diagnosis, and to encourage these networks to implement a series of management actions aimed at reducing the identified sources of vulnerability.

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